Big Data Assignment 2: Distributed Search Engine

By: Aleksandr Mikhailov AI-01

al.mikhailov@innopolis.university

Methodology

System Architecture

The system uses Docker Compose to deploy:

- **cluster-master**: Hadoop master (NameNode, ResourceManager) + Spark master
- **cluster-slave-1**: Hadoop worker (DataNode, NodeManager)
- **cassandra-server**: Cassandra/ScyllaDB for index storage

```
Parquet File → HDFS → MapReduce Indexer → Cassandra/ScyllaDB
                                                    ↑
User Query → Spark Ranker ──────────────────────────┘
```

Data Preparation

Implemented in `prepare_data.py`:

1. Read parquet file containing Wikipedia articles (id, title, text)
2. Sample 100 documents using PySpark
3. Save each document as `<doc_id>_<doc_title>.txt`
4. Transform to tab-separated format and store in HDFS at `/input/data`:

   ```
   <doc_id>\t<doc_title>\t<doc_text>
   ```

Indexing with Hadoop MapReduce

The design patter to implement the indexing is In-Mapper aggregation. This method was chosen because it significantly reduces the memory strain (at least on my system) in contrast to the standatd MapReduce approach.

Mapper (`mapper1.py`):

- Extracts terms using regex `[a-z0-9]+`
- Emits key-value pairs:
  - `DL:<doc_id>\t<length>` - document length
  - `DOCS_COUNT\t1` - document counter
  - `TF:<term>@<doc_id>\t<count>` - term frequency
  - `DF:<term>\t1` - document frequency
  - `TT:<doc_id>\t<title>` - document title (it is cleaned to include only ascii characters)

Reducer (`reducer1.py`):

- Aggregates values for each key
- Computes final TF, DF, DL, TT, and total document count

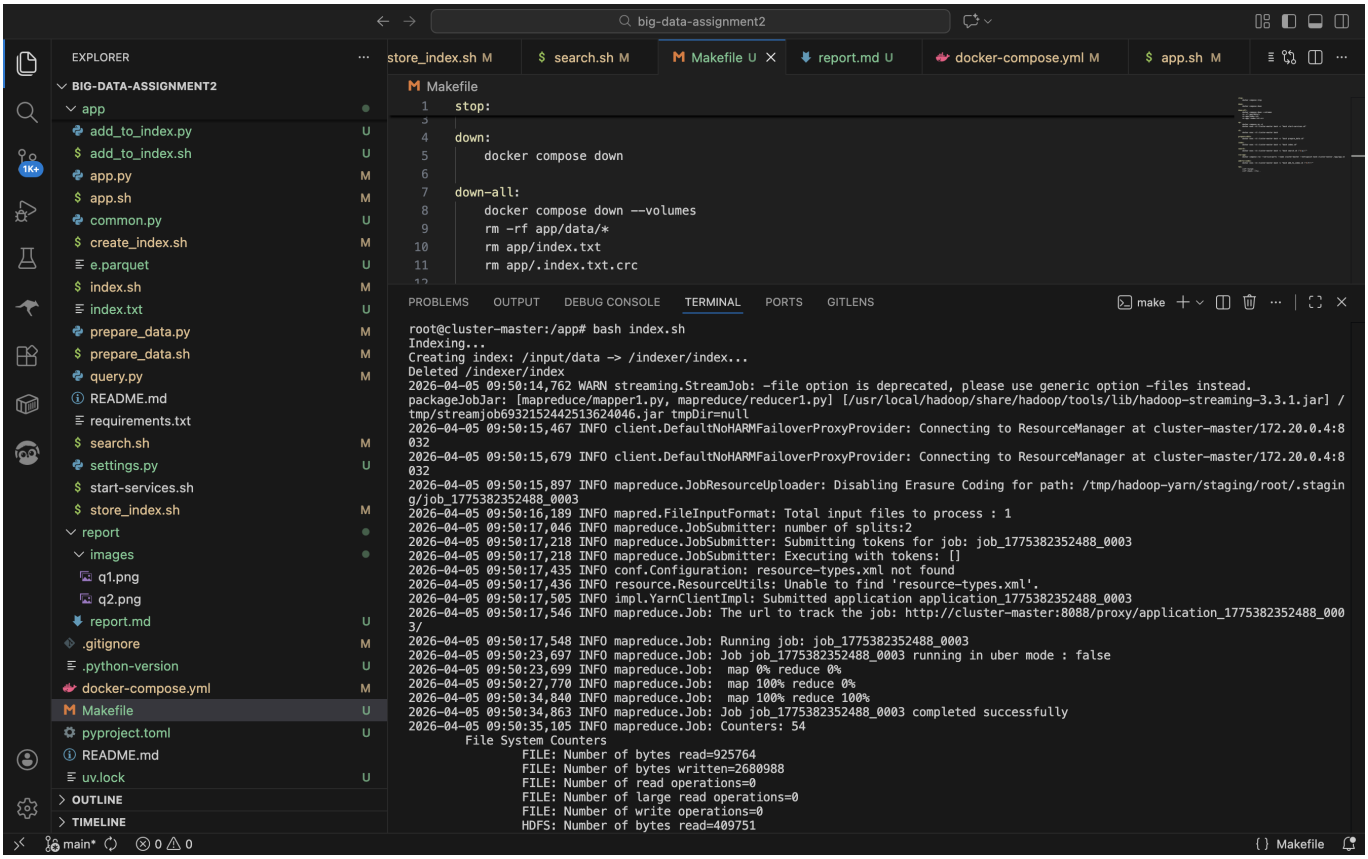
Index Storage in Cassandra/ScyllaDB

Implemented in `app.py`, creates four tables:

Table	Purpose	Primary Key
`global_data`	Total document count	`key`
`dl`	Document lengths	`doc_id`
`df`	Document frequencies	`term`
`tf`	Term frequencies	`(term, doc_id)`
`docs`	Document titles	`doc_id`

Data is loaded from merged MapReduce output using prepared statements with async inserts for performance.

Indexing Results



BM25 Ranking Algorithm

Implemented in `query.py`:

```
BM25(q,d) = Σ IDF(t) × (tf × (k1 + 1)) / (tf + k1 × (1 − b + b × (dl /
avg_dl)))
```

Parameters:

- `k1 = 1.0` (term frequency saturation)
- `b = 0.75` (length normalization)
- `IDF(t) = log(N / df(t))`

Query Processing

1. Parse query terms using regex
2. For each term, retrieve documents from Cassandra and compute BM25 scores
3. Aggregate scores per document using Spark RDD `reduceByKey`
4. Sort and return top 10 results

I have encountered a very frustratin issue with Cassandra server. It was constantly running out of memory and crashing, so I had to restart the containers or my machine just to run indexing and search.

Dynamic Index Updates

`add_to_index.py` allows adding new documents without full re-indexing:

- Updates total document count
- Adds document length
- Updates term and document frequencies

Documentation

Explore Makefile for the prepared scripts in my implementation. Mainly, they allow to run the node cluster, index the data, and run the search engine.

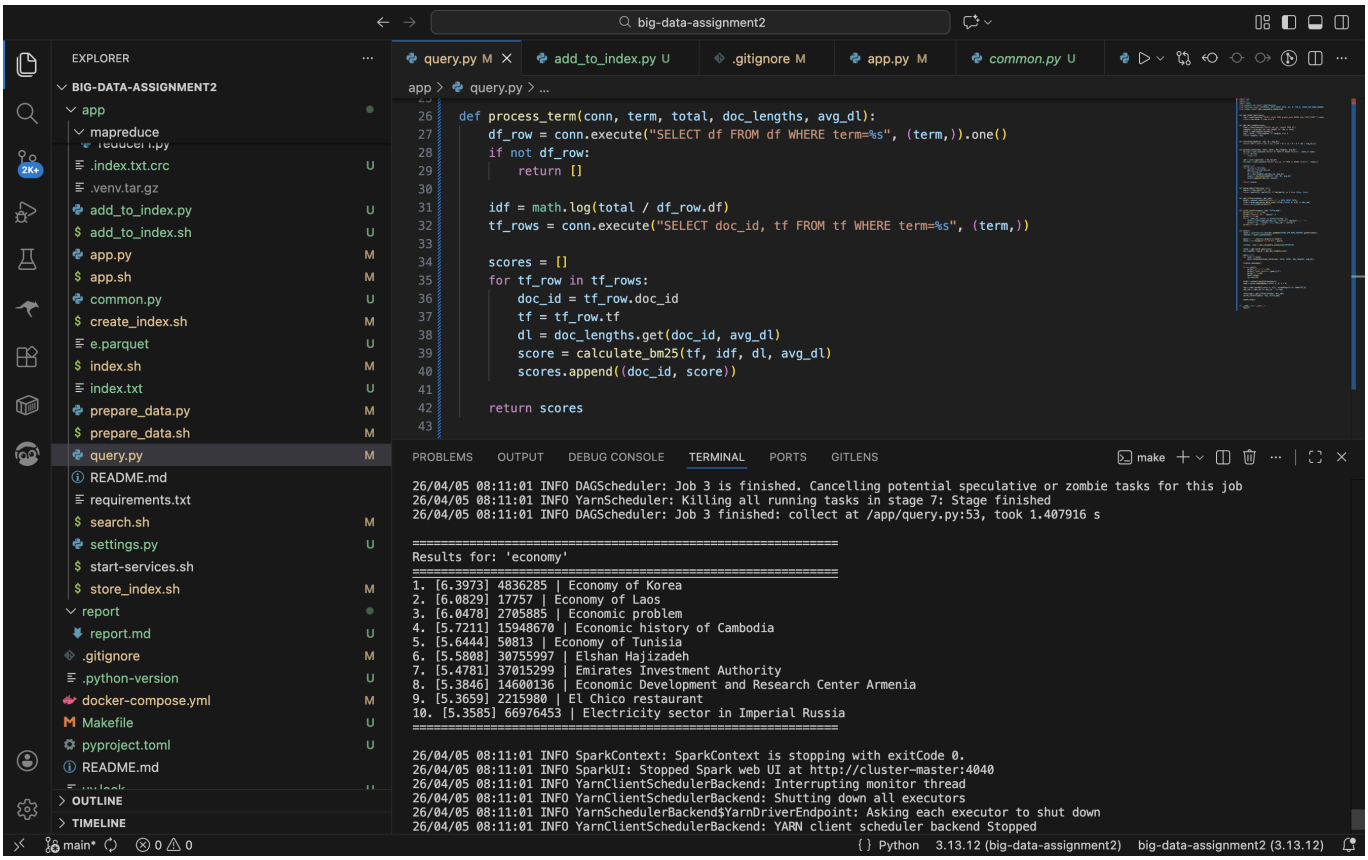
I have used `e.parquet` file, you can either manually download it to `app` directory or run `git lfs pull`, since it is tracked by lfs in this repo.

Configuration (app/settings.py)

Parameter	Value	Description
`SAMPLE_LIMIT`	1500	Documents to index
`TOP_K`	10	Results to return
`K1`	1.0	BM25 parameter
`B`	0.75	BM25 parameter
`KEYSPACE`	query_engine	Cassandra keyspace

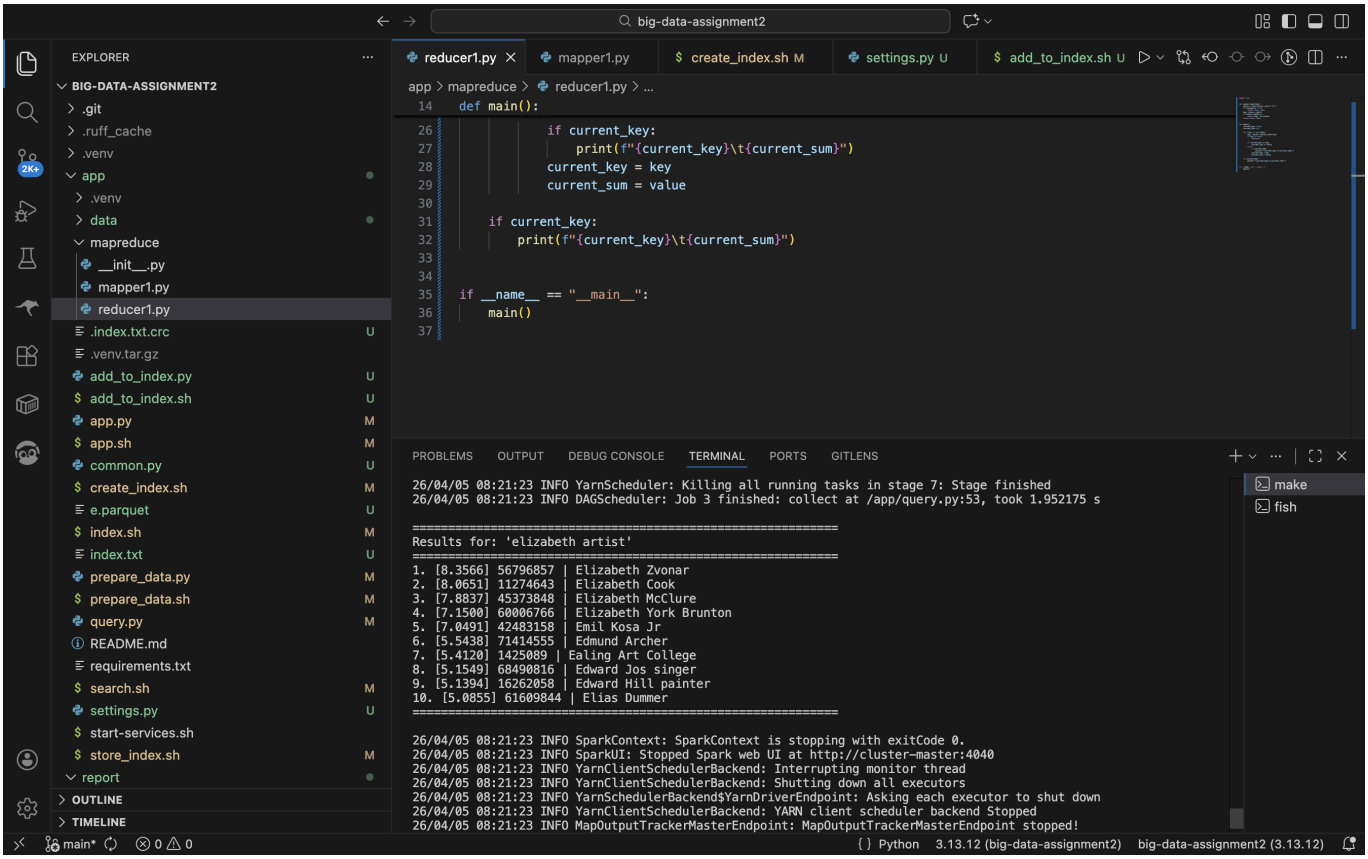
Query Demonstration

Query 1: "economy"



Analysis: Exact match about Korean economy receives highest score; related and similar economic documents follow.

Query 2: "elizabeth artist"



Analysis: Documents with "elizabeth" in title/content rank highest. Notably, the highest ranking entry is about Elizabeth Sonar, a Canadian contemporary artist.

Design Choices

- **Single pipeline**: Simple and sufficient for this scale
- **Tab-separated format**: Easy to parse, compatible with streaming
- **Separate Cassandra tables**: Optimizes query patterns
- **Async inserts**: Improves loading performance

Potential Improvements

- Add stemming/lemmatization for better term matching
- Implement stop word removal
- Add query result caching
- Shard Cassandra cluster for larger datasets

Challenges

- Docker cluster configuration required careful setup
- Data format consistency between components
- Python environment packaging for Spark YARN
- Cassandra server crashing with out of memory